SDEV2004 – Software for the Global Market

Continuous Assessment – Phase 1

Cultural and Usability Analysis and Design

Group 8 - Cross Paths	
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Important

The group report should present the shared conclusions and design decisions developed collaboratively by the group.

While the group's work may be informed by the individual cultural analyses carried out by group members, the group report should synthesise and summarise the group's collective findings rather than copying text from individual reports.

Similarly, the individual reports must present each student's own analysis and reasoning and should not simply reproduce sections of the group document.

Each student must work on a **substantive page that supports meaningful user interaction or information exploration**. It should involve more than a simple splash screen, login page, or static landing page.

The page should contain **multiple interface elements** (such as navigation, content sections, images, controls, or other components) that allow you to demonstrate **cultural adaptation in the interface design.**

1. Target Application Area

Description (approx. 150–200 words):

CrossPaths is a culturally adaptive speed-friending and social discovery web platform designed for international newcomers living in Dublin. The application allows users to discover, RSVP to, and attend structured social events – such as speed-friending sessions, language exchanges, and interest-based meetups – with the goal of building genuine friendships rather than professional networks. Unlike generic social platforms, CrossPaths adapts its user interface, information architecture, language tone, and visual design dynamically based on the user's selected cultural locale.

The primary users are recently relocated adults – professionals, students, and newcomers – who are seeking to establish a social circle in an unfamiliar city. The application targets culture markets three locales: Ireland (en_IE) as the default, Ukraine (uk_UA), and Brazil (pt_BR), representing three meaningfully different cultural profiles across Hofstede's Uncertainty Avoidance and Individualism dimensions.

This application area benefits strongly from cultural adaptation because the behaviours that drive social engagement – trust-building, risk tolerance, decision-making style, and communication preference – vary significantly across cultures. A single, culturally neutral design would systematically underserve users whose cultural norms differ from the assumed default. CrossPaths treats cultural adaptation not as a skin change, but as a fundamental difference in information architecture, navigation flow, visual tone, and copy register.

2. Cultural Scope

2.1 Cultural Dimensions Used

Identify the cultural dimensions used in your analysis.

Cultural Dimension	Explanation
Uncertainty Avoidance (UA)	Ireland: 35 (Low) Ukraine: 95 (Very High) Brazil: 76 (High)
Individualism vs Collectivism (IDV)	Ireland: 70 (High Indiv.) Ukraine: 25 (Collectivist) Brazil: 38 (Moderate)

Explain briefly why these dimensions are relevant to your application.

2.2 Selected Cultural Contexts

Your application must support **two countries and languages other than English.**

Country	Language	Cultural Characteristics
Ukraine	Ukrainian (uk_UA)	UA = 95 (Very High), IDV = 25 (Collectivist). Very low tolerance for ambiguity; strong need for verified authority, clear roles, and explicit rules. Trust earned slowly through institutional credentials. Communication is explicit, formal, and written. Decision-making is

		deliberate and research-driven. Social behaviour is cautious and small-group oriented.
Brazil	Portuguese (pt_BR)	UA = 76 (High), IDV = 38 (Moderate Collectivist). Trust built through warmth, shared humour, and mutual connections rather than formal credentials. Relationally expressive; communication is high-context and enthusiastic. Decision-making is spontaneous and social-proof driven. Prefers visual energy and social indicators over text density.

Explain:

- Ukraine and Brazil were selected because they represent two culturally distinct profiles that share a surface similarity (both have moderately high UA) but diverge sharply on IDV and on the type of certainty they seek. Ukraine represents a formal, credential-driven, detail-first profile; Brazil represents a relational, warmth-driven, social-proof-first profile. Both cultures are represented among Dublin's international newcomer population, making the contrast practically relevant.
- Ukraine (UA 95, IDV 25) requires high structural certainty and formal authority cues before social engagement. Brazil (UA 76, IDV 38) requires social certainty – warmth, relational trust, and social proof – rather than informational certainty. This contrast creates non-trivial, genuine design tensions that require real adaptation in layout, navigation, information architecture, and tone, not just translation.

3. User Understanding – Contrasting Personas

Describe the typical users of your application.

CrossPaths is designed for two primary user types, both of whom are international newcomers in Dublin seeking social connection. Their cultural backgrounds produce fundamentally different needs, behaviours, and expectations.

User Persona Type 1 – Oksana (Ukrainian / uk_UA profile)

Age 28. Recently relocated Ukrainian professional living in Dublin. Cultural profile: UA = 95 (Very High), IDV = 25 (Collectivist, risk-averse). Oksana builds trust slowly through verified credentials, official logos, and institutional backing. She expects clear roles and hierarchy – who is the host? what is the agenda? She has very low tolerance for ambiguity and needs exact event details (time, location, agenda, attendee count) before she will consider attending. She prefers explicit, formal, written instructions and dislikes vague or overly casual language. Her decision-making is deliberate: she researches all details before committing.

Goals:

1. find safe, structured meetup events;
2. know exactly what to expect before attending;
3. connect with Ukrainian speakers or culturally familiar people;
4. build a small, high-trust friend group.

Frustrations:

- events with no agenda feel unsafe;
- vague descriptions provide no usable information;
- no way to verify organisers;
- emoji-heavy interfaces feel untrustworthy.

Behaviour patterns:

- guidance over autonomy;
- detail over speed;
- structure over flexibility;
- low risk tolerance; formal digital behaviour.

User Persona Type 2 – Rafael (Brazilian / pt_BR profile)

Age 24. Recently arrived Brazilian newcomer in Dublin. Cultural profile: UA = 76 (High), IDV = 38 (Moderate collectivist, relational, expressive). Rafael builds trust through warmth, shared humour, and mutual social connections – social proof is his key trust signal. He is comfortable with flat, informal dynamics and moderate spontaneity. He communicates through emotion and enthusiasm, is emoji-friendly, and makes attendance decisions based on who else is going and the overall vibe.

Goals:

1. quickly find fun, lively events with other newcomers;
2. see who's attending and connect before the event;
3. express personality through his profile;
4. build a wide, diverse social network fast.

Frustrations:

- platforms that feel clinical or corporate;
- no way to see other attendees' personalities before committing;
- events too rigid or scripted;
- slow text-heavy onboarding.

Behaviour patterns:

- autonomy over guidance;
- speed over detail;
- flexibility over structure;
- high risk tolerance; expressive digital behaviour.

4. Point of View (PoV) Statements

2-4 PoV statements required.

Point of View (PoV) statements should be derived from the **user descriptions developed by the group**.

Each PoV should represent a **specific user type and a meaningful design problem** that the application should address.

Use the **Design Thinking PoV format:**

Template: User + Need + Insight

PoV 1 – Oksana (uk_UA) | Links to: Uncertainty Avoidance (UA = 95)

User: A recently relocated Ukrainian professional in Dublin – cautious, detail-oriented, and risk-averse, with a very high need for structure and verified authority before engaging socially.

Need: To feel fully informed and safe before committing to attend a social event with strangers.

Insight: Her very high uncertainty-avoidance cultural background (UA = 95) means that ambiguity, missing information, and unverified hosts do not merely inconvenience her – they erode trust entirely and prevent social engagement.

PoV 2 – Rafael (pt_BR) | Links to: Individualism vs. Collectivism (IDV = 38)

User: A recently arrived Brazilian newcomer in Dublin – expressive, socially-driven, and relational, who builds trust through warmth and shared social connections rather than formal credentials.

Need: To feel socially energised and personally connected through the platform before and during events.

Insight: His relationally-driven, moderate-collectivist cultural background (IDV = 38) means that impersonal, rigid, or visually flat interfaces suppress the social motivation he needs to engage – the platform itself must radiate warmth, not just facilitate logistics.

Each PoV statement must clearly **link to one of the cultural dimensions selected by the group**.

5. Hypotheses (Lean UX)

2-4 Hypotheses required.

Each hypothesis should be **derived from one of your Point of View (PoV) statements** and represent a proposed design response to the problem identified in that PoV.

Template:

We believe that implementing **[design change]** for **[user group]** will result in **[improved outcome]** because **[insight identified in the PoV].**

Hypothesis 1 – Derived from PoV 1 (Oksana / uk_UA)

We believe that implementing **verified host badges and institutional trust markers on every event card and event detail page** for **Ukrainian-locale users (uk_UA)** will result in **higher RSVP rates and reduced pre-event drop-off** because **Oksana's very high UA (95) means that unverified hosts are a hard blocker to social engagement, and visible credential signals directly reduce the ambiguity that prevents her from committing.**

Hypothesis 2 – Derived from PoV 1 (Oksana / uk_UA)

We believe that implementing **breadcrumb-style navigation and a step-by-step onboarding flow with explicit progress indicators** for **Ukrainian-locale users (uk_UA)** will result in **higher sign-up completion rates and lower abandonment** because **structured entry points with visible progress reduce the disorientation that unguided platforms create for users with very high uncertainty avoidance.**

Hypothesis 3 – Derived from PoV 2 (Rafael / pt_BR)

We believe that implementing **a visual-first event discovery feed that foregrounds attendee photos, personality tags, and social connection indicators** for **Brazilian-locale users (pt_BR)** will result in **higher event engagement and faster attendance decisions** because **Rafael's moderate collectivism (IDV = 38) means he makes social decisions based on who is going and whether the energy feels right, not on reading event credentials.**

Hypothesis 4 – Derived from PoV 1 and PoV 2 (Both locales)

We believe that implementing **muted blue/grey tones with formal, explicit copy for uk_UA and vibrant warm tones (yellow, orange, green) with enthusiastic emoji-inclusive copy for pt_BR** for **all locale-switching users** will result in **improved perceived platform credibility for Ukrainian users and improved emotional engagement for Brazilian users** because **surface-level cultural signalling (colour, tone) is the first cue users use to judge whether a platform was designed for them.**

6. Design and Development Guidelines

This section defines **shared rules the group will follow during development.**

These guidelines ensure **consistency, maintainability, and cultural coherence.**

6.1 Code and Project Structure

Define:

- **Flask folder structure:** crosspaths/ (root), crosspaths/app.py (entry point), crosspaths/config.py (locale list and flags), crosspaths/templates/ (all Jinja2 HTML templates), crosspaths/templates/base.html (shared base with navbar, footer, locale switcher), crosspaths/templates/partials/ (shared reusable components), crosspaths/static/css/ (shared stylesheet + per-locale overrides), crosspaths/static/img/ (assets organised by locale: /img/uk_UA/, /img/pt_BR/, /img/en_IE/), crosspaths/static/js/, crosspaths/translations/ (GNU gettext .po and .mo files), crosspaths/babel.cfg, requirements.txt.
- **Template inheritance strategy:** All pages extend base.html using Jinja2 {% extends 'base.html' %}. base.html contains the shared head, navigation bar with locale switcher, footer, and CSS/JS imports. No page-specific content belongs in base.html. Shared UI components (e.g. event cards) are extracted into {% include %} partials in templates/partials/ to avoid duplication.
- **Naming conventions:** Template files: lowercase_underscores (event_discovery.html). Python route functions: lowercase_underscores. CSS classes: kebab-case (.event-card). Translation strings: dot-notation keys ('event.rsvp_button'). Locale-specific static assets prefixed with locale code (uk_banner.jpg).
- **Rules the group has agreed:** No hardcoded language strings in templates – all user-visible text uses _() or gettext(). No hardcoded locale-specific pages – cultural adaptation via Jinja2 conditionals on get_locale(), not separate duplicate pages. No unused static assets committed. All members must test their page in at least two locales before merging. The group uses a shared Git repository with named feature branches per member (feature/kashish-discovery, feature/yana-profile, etc.) and merges via pull request.

6.2 Internationalisation and Localisation Strategy

Describe how translation will work.

Include:

- **Supported languages:** English/Ireland (en_IE) – default and fallback; Ukrainian (uk_UA) – full support; Portuguese/Brazil (pt_BR) – full support. All three locales receive both translation and cultural UI adaptation.
- **Language switching behaviour:** A locale switcher is present in the shared navigation bar (base.html) on every page. The active locale is stored in the user's session (session['locale']) and retrieved via a get_locale() function passed to Flask-Babel. Switching locale reloads the current page in the new locale. Default is en_IE; unsupported locales fall back silently to en_IE.
- **Translation approach:** GNU gettext via Flask-Babel. All user-visible strings in Python files wrapped with gettext() or _(). All user-visible strings in Jinja2 templates wrapped with {{ _('string') }} or {% trans %}...{% endtrans %} for multi-line blocks. Translations extracted with pybabel extract, compiled with pybabel compile. Each locale has translations/[locale]/LC_MESSAGES/messages.po. Compiled .mo files are committed to the

repository. No member may add a user-visible string without simultaneously adding it to all three .po files.

6.3 Cultural Adaptation Guidelines

Explain **what UI elements will adapt culturally.**

Possible adaptations:

Element	**Adaptation**
Layout	uk_UA: Structured grid, clear labelled sections, high information density, explicit CTAs at every stage. pt_BR: Warm social feed, layered visual cards, attendee grid prominent. | UA dimension | H1, H3, H4
Navigation	uk_UA: Step-by-step breadcrumb navigation, linear flows, explicit progress indicators. pt_BR: Browse-first, explorative, swipe-friendly card stack. | UA dimension | H2
Information structure / content density	uk_UA: Full event details (agenda, host bio, attendee cap, location) surfaced at discovery stage. pt_BR: Minimal text on cards; lead with photos, attendee count, and social indicators. | UA + IDV | H1, H3
Colour	uk_UA: Muted blue and grey palette – trustworthy, formal, minimal decoration. pt_BR: Vibrant warm tones (yellow, orange, green) – energetic and welcoming. | UA + IDV | H4
Images	uk_UA: Verified host headshots, venue photos, safety/credential icons. pt_BR: Personality photos, group energy shots, expressive emoji reactions. | IDV dimension | H1, H3
Symbols	uk_UA: Trust badges, verification checkmarks, structured agenda icons. pt_BR: Emoji reactions, mutual connection icons, group energy indicators. Copy tone: uk_UA formal and explicit (no emojis); pt_BR warm, conversational, emoji-inclusive. | UA + IDV | H1, H4

Describe **important design trade-offs or discussions.**

Examples:

- Why a particular colour scheme was chosen
- Why certain cultural adaptations were rejected
- How usability concerns influenced design decisions

6.4 Individual Contribution Plan

Define which student will implement which page.

Student Name	Planned Page
Yana Chulovska	Home page
Lívia Guimarães	Events page
Kashish Kakran	Communities page
Simphiwe Klassen	Login page

Important

Each student must work on a **substantive page that supports meaningful user interaction or information exploration**. It should involve more than a simple splash screen, login page, or static landing page.

The page should contain **multiple interface elements** (such as navigation, content sections, images, controls, or other components) that allow you to demonstrate **cultural adaptation in the interface design.**

6.5 Integration plan

Explain how individual pages will be integrated into the shared Flask application (e.g., template inheritance, shared navigation, shared language switching logic).

All individual pages extend the shared base.html template. The base template handles the locale switcher, navigation bar, and CSS variable injection based on the active locale. Cultural CSS variables (colour palette, font weight, spacing) are defined in a locale_theme() context processor in app.py and injected into every template via Jinja2 context – no member needs to manually check the locale for colour changes. Locale-conditional content blocks use {% if locale == 'uk_UA' %}...{% elif locale == 'pt_BR' %}...{% else %}...{% endif %} within each member's template.

Explain how you will ensure that the application works on more than one machine.

requirements.txt is maintained by all members – anyone adding a new package must update it immediately. No absolute file paths are used; all static asset references use Flask's url_for('static', filename='...') helper. Environment-specific settings are stored in a .env file listed in .gitignore; a .env.example with placeholder values is committed instead. Before the Phase 2 submission, each member must clone the repository on a fresh machine and verify the application runs correctly with pip install -r requirements.txt followed by flask run. The group README will include clear setup and run instructions as required by the specification.

7. UI Sketches / Wireframes (Optional – delete if not using)

Include:

- Page sketches
- Layout ideas

- Navigation flow

These can be:

- Hand-drawn
- Sketches
- Visio
- Figma
- Wireframe tools